

2026-05-07-p2-f28-kilocode-refactoring-design

Phase 2 / Feature 28 — Kilo-code AST-Aware Refactoring

Date: 2026-05-07 | **Status:** Approved **Programme:** CLI-Agent Fusion — Phase 2 port (kilo-code)

1. Goal

Ship AST-aware multi-file refactoring: cross-file rename, semantic multi-edit blocks, codebase impact analysis with call-graph + blast radius, and refactoring utilities (extract method, move, inline). Reuses F17 smart-edit for atomic file edits, repomap/tree-sitter for AST parsing and call graph construction.

2. Architecture

internal/kilocode/ (NEW) — 3 sub-components: - **RenameEngine**: tree-sitter query all references → atomic F17 edits → verify - **ImpactAnalyzer**: call graph from tree-sitter → affected files/functions → blast radius - **Refactorer**: extract method, move symbol, inline call (AST transforms)

Tools: kilocode_rename, kilocode_impact, kilocode_multi_edit. Slash: /kilocode.

3. Task Breakdown (8 tasks)

#	Task	Description
T01	Bootstrap	F28 evidence + PROGRESS + CONTINUATION
T02	types.go + callgraph.go	CallGraph + SymbolRef + sentinels (TDD)
T03	rename.go	Tree-sitter query + atomic F17 edits (TDD)
T04	impact.go	ImpactAnalyzer with call graph + blast radius (TDD)
T05	refactor.go	Extract method, move, inline (TDD)
T06	kilocode_tools.go	3 tools + /kilocode slash (TDD)
T07	main.go wiring	Register tools + slash
T08	Challenge harness + close-out	5 phases + push 4 remotes

Zero new deps — tree-sitter already indirect, F17/F25/repomap already shipped.